Operations Management - II

Term Project: Integrated Operations Management Business Case Analysis

Product: Skoda Kushaq (Compact SUV)

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Company and Product Overview

Skoda Auto Volkswagen India Pvt. Ltd. builds the Skoda Kushaq, a compact SUV, at its Chakan plant near Pune, Maharashtra, for the Indian domestic market. For this report, we act as an internal Operations Management team asked to put together an 8-month production plan and a component-level material plan for the Kushaq, and to design an inventory policy for one of its key purchased parts.

A few assumptions set the scope for what follows:

- The planning horizon is 8 months which gives the multi-level material plan enough room to settle into a steady rhythm once the initial pipeline is filled.
- The Kushaq is built through a 4-level product structure: the vehicle itself, 3 major sub-assemblies which are - engine, chassis, body, and the parts and sub-parts under each.
- All costs are illustrative and quoted in INR to show how the calculation works, not to represent Skoda's actual cost structure.
- Workforce and regular-time capacity stay flat across the horizon. Overtime and subcontracting soak up demand peaks, and a short backlog is allowed mid-plan as long as it clears to 0 by the last period, per the hybrid strategy the brief calls for.

Problem Scenario 1: Aggregate Planning and Material Requirements Planning

Demand Forecast

Monthly demand for the Kushaq starts at 1,000 units in Month 1, climbs to a peak of 2,200 units in Month 6, then eases back to 1,300 units by Month 8. Total demand across the 8 months comes to 12,500 units. This forecast drives both the Aggregate Production Plan and, downstream, the Master Production Schedule that feeds the MRP.

Aggregate Production Plan (Hybrid Strategy)

The plan uses a hybrid strategy that blends level and chase elements. Regular-time output stays constant across all 8 months, the level part, with no hiring or layoffs. Overtime and subcontracting then chase whatever demand regular time and on-hand stock can't cover, up to set limits. If a shortfall still remains after both, it carries forward as a short-term backlog, which has to clear fully by the end of the 8-month cycle.

- Regular-time production rate (RP): 1,350 units a month, held constant
- Overtime cap (OT): 250 units a month
- Subcontracting cap (SC): 300 units a month
- Backlog: allowed mid-plan, but must be zero by the end of Month 8

The plan is worked out month by month using this logic:

Available(t) = Beginning Inventory(t) + RP(t)

If Available(t) < Demand(t): Overtime(t) = min (OT_MAX, Shortfall)

If still short after OT: Subcontract(t) = min (SC_MAX, Remaining shortfall)

Ending Inventory/(Backlog)(t) = Beginning Inv.(t) + RP(t) + OT(t) + SC(t) − Demand(t)

Running this month by month, inventory builds steadily from Month 1 through Month 4 while demand sits below capacity. Overtime kicks in first, then subcontracting, as demand climbs through Months 5 and 6. By the end of Month 6, demand of 2,200 units outruns even the combined overtime and subcontract ceiling, leaving a backlog of 300 units. That backlog is fully worked off by Month 8, so the plan ends exactly where it needs to: 0 backlog.

Aggregate Plan Cost

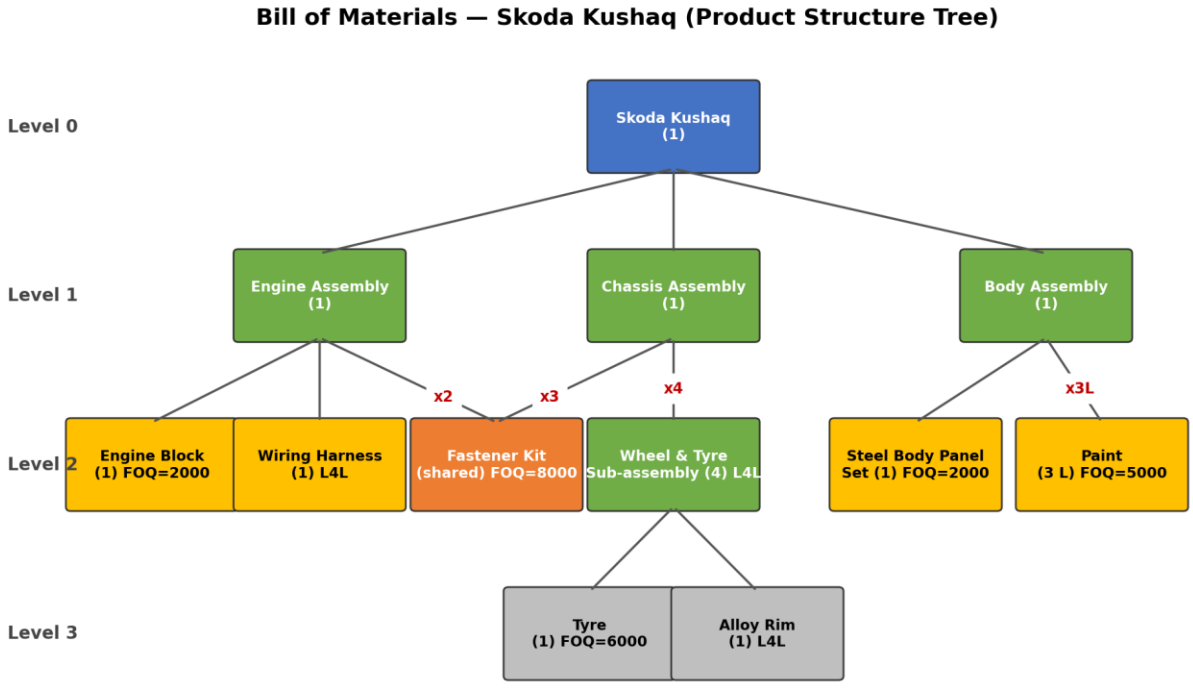
Total Cost = (RP × Regular-time cost) + (Σ OT × Overtime cost) + (Σ SC × Subcontract cost)

+ (Σ positive ending inventory × Holding cost) + (Σ backlog × Backorder cost)

Total Aggregate Plan Cost comes to about Rs 887.60 Crore.

Bill of Materials (BOM)

The Kushaq's product structure runs 4 levels deep. One part, the Fastener Kit, shows up under two different sub-assemblies (Engine Assembly and Chassis Assembly), each at a different quantity per parent. That satisfies the requirement that at least one component appear in more than one place in the structure. Lot-sizing is deliberately mixed: assemblies built close to exact need use Lot-for-Lot (L4L), while purchased components ordered in economical batches use Fixed-Order-Quantity (FOQ). Three components carry on-hand inventory (Engine Block, Fastener Kit, and Steel Body Panel Set), and every item in the structure has a defined lead time.



Product Structure Tree - Skoda Kushaq

From Aggregate Plan to Master Production Schedule (MPS)

The Master Production Schedule spells out how many finished Kushaq units need to be completed each month. It's taken straight from the Total Supply figure in the Aggregate Production Plan, regular time plus overtime plus subcontract units, since that's what's actually built each month, not just what customers ask for.

MRP Explosion Logic

Every item in the BOM runs through the same 5-step logic, applied top-down through the product structure:

Gross Requirement(t) = MPS(t) [top level]

OR

Σ (Parent's Planned Order Release(t) × Qty-per-parent) [lower levels]

Projected Available Balance(t) = Balance(t−1) + Planned Order Receipt(t) − Gross Requirement(t)

Net Requirement(t) = Gross Requirement(t) − Balance(t−1)

only if Balance(t−1) < Gross Requirement(t)

Planned Order Receipt(t):

L4L → exactly Net Requirement(t);

FOQ → smallest multiple of the lot size ≥ Net Requirement(t)

Planned Order Release = Planned Order Receipt, shifted earlier by the item's Lead Time

MRP Results and What the Matrices Show

The full MRP runs for all 12 items in the structure: the finished vehicle, the 3 sub-assemblies, and the 8 components and sub-components beneath them. Each item gets its own 8-month row for Gross Requirements, Projected Available Balance, Net Requirements, Planned Order Receipts, and Planned Order Releases. A few results stand out.

- The Fastener Kit's Gross Requirement each month is the sum of 2 separate demand streams, 2 kits per Engine Assembly release plus 3 kits per Chassis Assembly release. That confirms the shared-component logic is actually flowing through correctly.
- FOQ items (Engine Block, Fastener Kit, Steel Body Panel Set, Paint, Tyre) show the step pattern you'd expect: Planned Order Receipts always land on a multiple of the lot size. L4L items, by contrast, order exactly what's needed each month, no more.
- The 3 items carrying on-hand inventory show a Projected Available Balance that fluctuates, since lot-size rounding leaves small surpluses that get drawn down over the following months. Every zero-on-hand, lot-for-lot item, by comparison, settles at a flat balance of 0.

Problem Scenario 2: Inventory Management - Tyre

The Tyre gets its own dedicated inventory policy. It's high-volume, moderate-cost, and consumed steadily (4 per vehicle), and running out of it would stop the assembly line outright. That combination makes it a strong candidate for a formal reorder policy, separate from the MRP-driven batch shown in Aggregate Plan.

Model Choice: Continuous Review (Q-Model)

A continuous review (Q-model) makes more sense here than a periodic review (P-model), for 3 reasons specific to this part:

- Tyres move fast and steadily off the shelf, and the plant's warehouse system already tracks stock continuously, so a perpetual reorder trigger costs nothing extra to run.
- A tyre stock-out halts final assembly, an expensive line-down event. Continuous monitoring catches a shortfall the moment it crosses the reorder point, rather than waiting for the next scheduled review as a P-model would.
- Demand here is fairly stable month to month, so placing the same cost-minimizing quantity (EOQ) every time stock hits the reorder point is simpler to run than recalculating a variable order-up-to amount at each periodic review.

Assumptions for the Tyre Inventory Policy

- Mean monthly demand ($\bar{D}$): 6,600 tyres a month
- Standard deviation of monthly demand (σ): 900 tyres a month
- Supplier lead time (LT): 2 months
- Ordering/setup cost (S): Rs 30,000 per order
- Holding cost (H): Rs 60 per tyre per month
- Target cycle service level: 95% (Z = 1.645, from the standard normal table)

Economic Order Quantity (EOQ)

$$Q^* = \sqrt{(2 \times \bar{D} \times S) / H}$$

$Q^* = \sqrt{(2 \times 6{,}600 \times 30{,}000) / 60} = \sqrt{6{,}600{,}000} = 2{,}569$ tyres ~ 2,570 tyres per order

Safety Stock and Reorder Point (ROP)

$$\sigma_{LT} = \sigma \times \sqrt{LT}$$

$\sigma_{LT} = 900 \times \sqrt{2} = 1{,}272.8$ tyres

$$\text{Safety Stock (SS)} = Z \times \sigma_{LT}$$

$SS = 1.645 \times 1{,}272.8 = 2{,}094$ tyres, ~ 2,090 tyres.

$$\text{Reorder Point (ROP)} = (\bar{D} \times LT) + \text{Safety Stock}$$

$ROP = (6{,}600 \times 2) + 2{,}094 = 13{,}200 + 2{,}094 = 15{,}294$ tyres ~ 15,290 tyres.

Resulting Policy

Under this policy, the plant places a fixed order of about 2,570 tyres whenever on-hand stock, plus anything already on order, drops to 15,290 units. The safety stock of about 2,090 tyres cushions against demand swings during the 2-month replenishment lead time, sized to avoid a stock-out 95% of the time. That works out to roughly 30.8 orders a year, or about one every 12 days.

Conclusion

This case shows how the 3 planning layers at Skoda Auto Volkswagen India connect. The Aggregate Production Plan turns an 8-month demand forecast into a level regular-time output plan, topped up by overtime and subcontracting, which keeps total cost in check while allowing a short backlog that fully recovers, rather than forcing expensive permanent capacity changes. That plan's monthly output becomes the Master Production Schedule, which gets exploded through a 4-level Bill of Materials, mixing lot-for-lot and fixed-order-quantity rules, to produce the planned orders for every component. The Fastener Kit shows this mechanism at its clearest: it's genuinely shared across 2 different sub-assemblies, and its numbers reflect both. Finally, the Tyre gets pulled out for its own continuous-review inventory policy, because a part this critical to the assembly line deserves its own reorder-point discipline rather than riding on the batch logic already built into the MRP run.

Assumptions and Sources

The project brief tells us to “make necessary assumptions regarding the product, geographical location, customer details, and other variables,” and therefore, none of the figures above or in excel come from Skoda's actual data. They were built to be internally consistent so the aggregate planning, MRP, and inventory-policy calculations would all work end to end. It mixes a handful of verified real-world facts with a much larger set of illustrative assumptions built to make the calculations hang together. The two are kept separate so it's clear which figures are sourced and which are assumed for this term project.
The only use of AI in this report was to construct the Product Structure Tree - Skoda Kushaq diagram.

Aggregate Production Plan (Hybrid Strategy) - Skoda Kushaq

Parameters	
Regular-time production (RP, units/month)	1350
Overtime cap (OT_MAX, units/month)	250
Subcontract cap (SC_MAX, units/month)	300
Regular-time cost (Rs/unit)	700000
Overtime cost (Rs/unit)	750000
Subcontract cost (Rs/unit)	780000
Holding cost (Rs/unit/month)	5000
Backorder cost (Rs/unit/month)	15000

Month-by-month plan

Month	Demand	Beg. Inv/(Backlog)	Regular Prod (RP)	Overtime (OT)	Subcontract (SC)	Total Supply	End Inv/(Backlog)
1	1000	0	1350	0	0	1350	350
2	1100	350	1350	0	0	1350	600
3	1300	600	1350	0	0	1350	650
4	1700	650	1350	0	0	1350	300
5	2100	300	1350	250	200	1800	0
6	2200	0	1350	250	300	1900	-300
7	1800	-300	1350	250	300	1900	-200
8	1300	-200	1350	150	0	1500	0

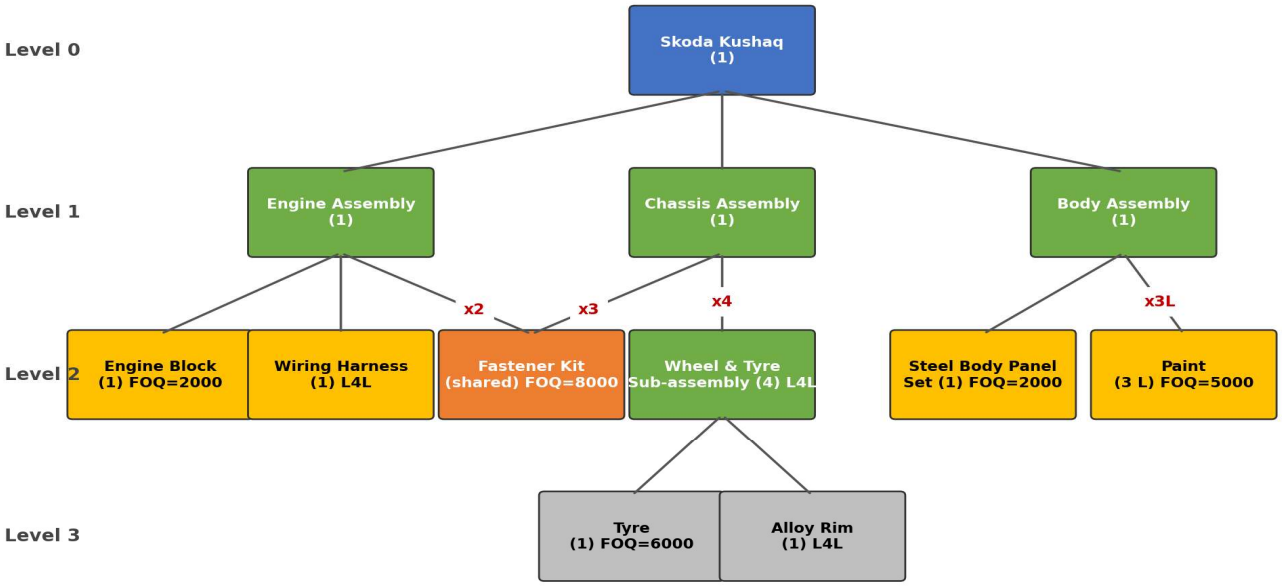
Cost Breakdown

Cost component	Rate	Quantity (formula)	Cost (Rs)
Regular time	700000	10800	7560000000
Overtime	750000	900	675000000
Subcontract	780000	800	624000000
Inventory holding	5000	1900	9500000
Backorder	15000	500	7500000
TOTAL APP COST (Rs)			**8876000000**

Bill of Materials - Skoda Kushaq

Item	Level	Parent	Qty/Parent	Lot rule	Lot size	Lead time (mo.)	On-hand inv.
Skoda Kushaq (finished vehicle)	0	—	1	L4L	—	1	15
Engine Assembly	1	Kushaq	1	L4L	—	1	0
Chassis Assembly	1	Kushaq	1	L4L	—	1	0
Body Assembly	1	Kushaq	1	L4L	—	1	0
Engine Block	2	Engine Assembly	1	FOQ	2000	2	1000
Wiring Harness	2	Engine Assembly	1	L4L	—	1	0
Fastener Kit	2	Engine Asm.(x2)/Chassis Asm.(x3)	2 / 3	FOQ	8000	1	6000
Wheel & Tyre Sub-assembly	2	Chassis Assembly	4	L4L	—	1	0
Steel Body Panel Set	2	Body Assembly	1	FOQ	2000	2	1000
Paint (litres)	2	Body Assembly	3	FOQ	5000	1	0
Tyre	3	Wheel & Tyre Sub-asm.	1	FOQ	6000	2	0
Alloy Rim	3	Wheel & Tyre Sub-asm.	1	L4L	—	1	0

Bill of Materials — Skoda Kushaq (Product Structure Tree)



Material Requirements Planning - Skoda Kushaq (all items, Months 1 - 8)

		Period	M1	M2	M3	M4	M5	M6	M7	M8
Item:	**Skoda Kushaq (finished vehicle)**									
OH:	15									
LT (mo):	1									
Lot rule:	L4L									
		Gross Requirements	1350	1350	1350	1350	1800	1900	1900	1500
		Projected Avail. Balance	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
		Net Requirements	1335	1350	1350	1350	1800	1900	1900	1500
		Planned Order Receipts	1335	1350	1350	1350	1800	1900	1900	1500
		Planned Order Releases	2685	1350	1350	1800	1900	1900	1500	0
Item:	**Engine Assembly**									
OH:	0									
LT (mo):	1									
Lot rule:	L4L									
		Gross Requirements	2685	1350	1350	1800	1900	1900	1500	0
		Projected Avail. Balance	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
		Net Requirements	2685	1350	1350	1800	1900	1900	1500	0
		Planned Order Receipts	2685	1350	1350	1800	1900	1900	1500	0
		Planned Order Releases	4035	1350	1800	1900	1900	1500	0	0
Item:	**Chassis Assembly**									
OH:	0									
LT (mo):	1									
Lot rule:	L4L									
		Gross Requirements	2685	1350	1350	1800	1900	1900	1500	0
		Projected Avail. Balance	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
		Net Requirements	2685	1350	1350	1800	1900	1900	1500	0
		Planned Order Receipts	2685	1350	1350	1800	1900	1900	1500	0
		Planned Order Releases	4035	1350	1800	1900	1900	1500	0	0
Item:	**Body Assembly**									
OH:	0									
LT (mo):	1									
Lot rule:	L4L									
		Gross Requirements	2685	1350	1350	1800	1900	1900	1500	0
		Projected Avail. Balance	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
		Net Requirements	2685	1350	1350	1800	1900	1900	1500	0
		Planned Order Receipts	2685	1350	1350	1800	1900	1900	1500	0
		Planned Order Releases	4035	1350	1800	1900	1900	1500	0	0
Item:	**Engine Block**									
OH:	1000									
LT (mo):	2									

Lot rule:	FOQ	2000								
		Gross Requirements	4035	1350	1800	1900	1900	1500	0	0
		Projected Avail. Balance	**965**	**1615**	**1815**	**1915**	**15**	**515**	**515**	**515**
		Net Requirements	3035	385	185	85	0	1485	0	0
		Planned Order Receipts	4000	2000	2000	2000	0	2000	0	0
		Planned Order Releases	8000	2000	0	2000	0	0	0	0
Item:	**Wiring Harness**									
OH:	0									
LT (mo):	1									
Lot rule:	L4L									
		Gross Requirements	4035	1350	1800	1900	1900	1500	0	0
		Projected Avail. Balance	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
		Net Requirements	4035	1350	1800	1900	1900	1500	0	0
		Planned Order Receipts	4035	1350	1800	1900	1900	1500	0	0
		Planned Order Releases	5385	1800	1900	1900	1500	0	0	0
Item:	**Fastener Kit**									
OH:	6000									
LT (mo):	1									
Lot rule:	FOQ	8000								
		Gross Requirements	20175	6750	9000	9500	9500	7500	0	0
		Projected Avail. Balance	**1825**	**3075**	**2075**	**575**	**7075**	**7575**	**7575**	**7575**
		Net Requirements	14175	4925	5925	7425	8925	425	0	0
		Planned Order Receipts	16000	8000	8000	8000	16000	8000	0	0
		Planned Order Releases	24000	8000	8000	16000	8000	0	0	0
Item:	**Wheel & Tyre Sub-assembly**									
OH:	0									
LT (mo):	1									
Lot rule:	L4L									
		Gross Requirements	16140	5400	7200	7600	7600	6000	0	0
		Projected Avail. Balance	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
		Net Requirements	16140	5400	7200	7600	7600	6000	0	0
		Planned Order Receipts	16140	5400	7200	7600	7600	6000	0	0
		Planned Order Releases	21540	7200	7600	7600	6000	0	0	0
Item:	**Steel Body Panel Set**									
OH:	1000									
LT (mo):	2									
Lot rule:	FOQ	2000								
		Gross Requirements	4035	1350	1800	1900	1900	1500	0	0
		Projected Avail. Balance	**965**	**1615**	**1815**	**1915**	**15**	**515**	**515**	**515**
		Net Requirements	3035	385	185	85	0	1485	0	0
		Planned Order Receipts	4000	2000	2000	2000	0	2000	0	0
		Planned Order Releases	8000	2000	0	2000	0	0	0	0

Item:	**Paint (litres)**									
OH:	0									
LT (mo):	1									
Lot rule:	FOQ	5000								
		Gross Requirements	12105	4050	5400	5700	5700	4500	0	0
		Projected Avail. Balance	**2895**	**3845**	**3445**	**2745**	**2045**	**2545**	**2545**	**2545**
		Net Requirements	12105	1155	1555	2255	2955	2455	0	0
		Planned Order Receipts	15000	5000	5000	5000	5000	5000	0	0
		Planned Order Releases	20000	5000	5000	5000	5000	0	0	0
Item:	**Tyre**									
OH:	0									
LT (mo):	2									
Lot rule:	FOQ	6000								
		Gross Requirements	21540	7200	7600	7600	6000	0	0	0
		Projected Avail. Balance	**2460**	**1260**	**5660**	**4060**	**4060**	**4060**	**4060**	**4060**
		Net Requirements	21540	4740	6340	1940	1940	0	0	0
		Planned Order Receipts	24000	6000	12000	6000	6000	0	0	0
		Planned Order Releases	42000	6000	6000	0	0	0	0	0
Item:	**Alloy Rim**									
OH:	0									
LT (mo):	1									
Lot rule:	L4L									
		Gross Requirements	21540	7200	7600	7600	6000	0	0	0
		Projected Avail. Balance	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
		Net Requirements	21540	7200	7600	7600	6000	0	0	0
		Planned Order Receipts	21540	7200	7600	7600	6000	0	0	0
		Planned Order Releases	28740	7600	7600	6000	0	0	0	0

Continuous Review (Q-Model) Inventory Policy - Tyre

Inputs (assumptions)	
Mean monthly demand, $\bar{D}$ (tyres/month)	6600
Std. dev. of monthly demand, σ (tyres/month)	900
Supplier lead time, LT (months)	2
Ordering/setup cost, S (Rs/order)	30000
Holding cost, H (Rs/tyre/month)	60
Cycle service level (Z-value)	1.645

Formulas and results	
Economic Order Quantity, $Q^* = \sqrt{(2 \times \bar{D} \times S / H)}$	**2,569.0**
Std. dev. of demand during lead time, $\sigma_LT = \sigma \times \sqrt{LT}$	**1,272.8**
Safety Stock, $SS = Z \times \sigma_LT$	**2,093.7**
Reorder Point, $ROP = (\bar{D} \times LT) + SS$	**15,293.7**
Orders per year = $(\bar{D} \times 12) / Q^*$	**30.8**
Cycle length (days) = 365 / Orders per year	**11.8**

Resulting Policy Summary	
Parameter	**Value**
Order quantity (Q*, EOQ, rounded)	2570
Safety stock (SS, rounded)	2090
Reorder point (ROP, rounded)	15290
Approx. orders per year	30.8285582
Approx. cycle length (days)	11.8396714